Bias-Variance

Recap

- Previously:
  - VC-dimension:
    - $\Pr[|E_{in}(h^*) - E_{out}(h^*)| > \varepsilon] \leq 8((2n)^d+1)e^{-\varepsilon^2 n/8}$
    - Proved via union bound, i.e. actually prove:
    - $\Pr[\exists\, h : |E_{in}(h) - E_{out}(h)| > \varepsilon] \leq 8((2n)^d+1)e^{-\varepsilon^2 n/8}$
    - Probability is over the random sample.
- This time:
  - Bias-Variance.
    - Alternative bound on E_{out}.
    - Inspires several algorithms:
      - Bagging
      - Random Forest

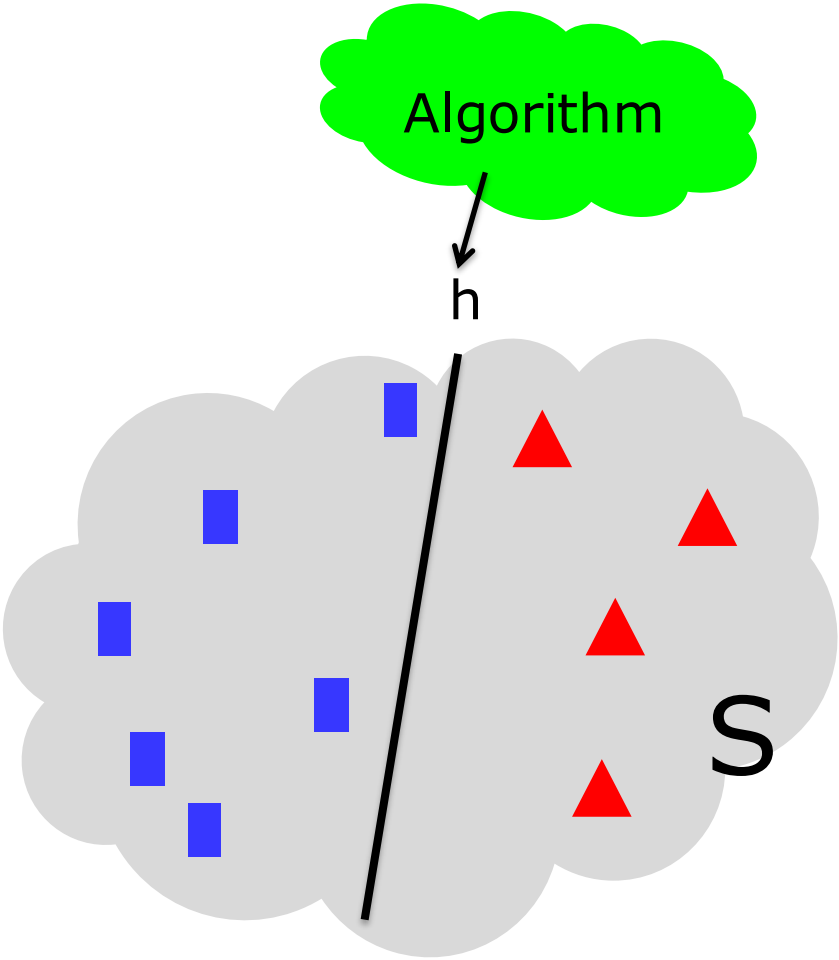
Bias Variance

- Different loss function:
  - Least squares loss:
    - $E_{out}(h) = \mathbb{E}_x[(h(x)-f(x))^2]$
    - $E_{in}(h) = \frac{1}{n}\sum_{i=1}^{n}(h(x_i) - f(x_i))^2$
  - Here x is drawn from some unknown distribution.
  - Training data consists of n i.i.d. samples from same distribution.

- Note: For labels in {0,1}, squared loss equals 0-1 loss.
- For labels in {-1,+1}, squared loss equals 4 x 0-1 loss.
- Implication: Bias-Variance also makes sense for 0-1 loss.

Bias Variance

- Least squares loss:
  - $E_{out}(h) = \mathbb{E}_x[(h(x)-f(x))^2]$
  - $E_{in}(h) = \frac{1}{n}\sum_{i=1}^{n}(h(x_i) - f(x_i))^2$

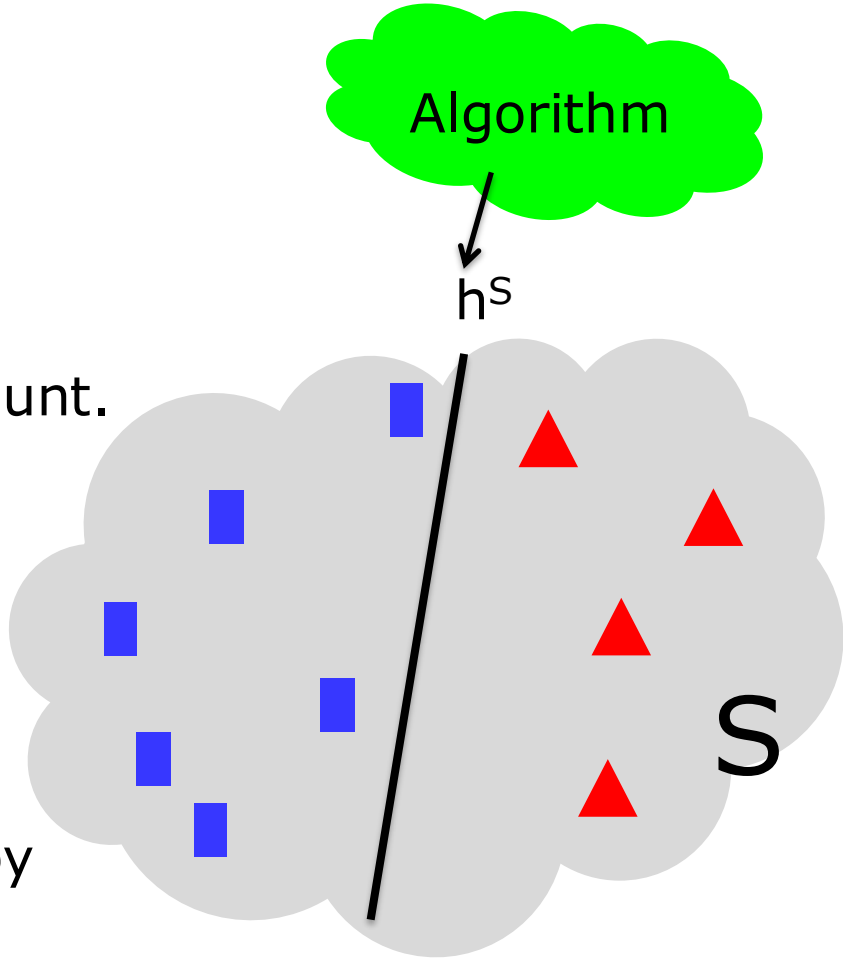


- Learning Setup:
  - Receive n samples S i.i.d. from unknown distribution.
  - Run learning algorithm to get hypothesis h.

Bias Variance

- Bias-Variance:
  - Bound E_{out}.
  - Take learning algorithm into account.

- Formally we bound:
  - $\mathbb{E}_S[E_{out}(h^S)]$

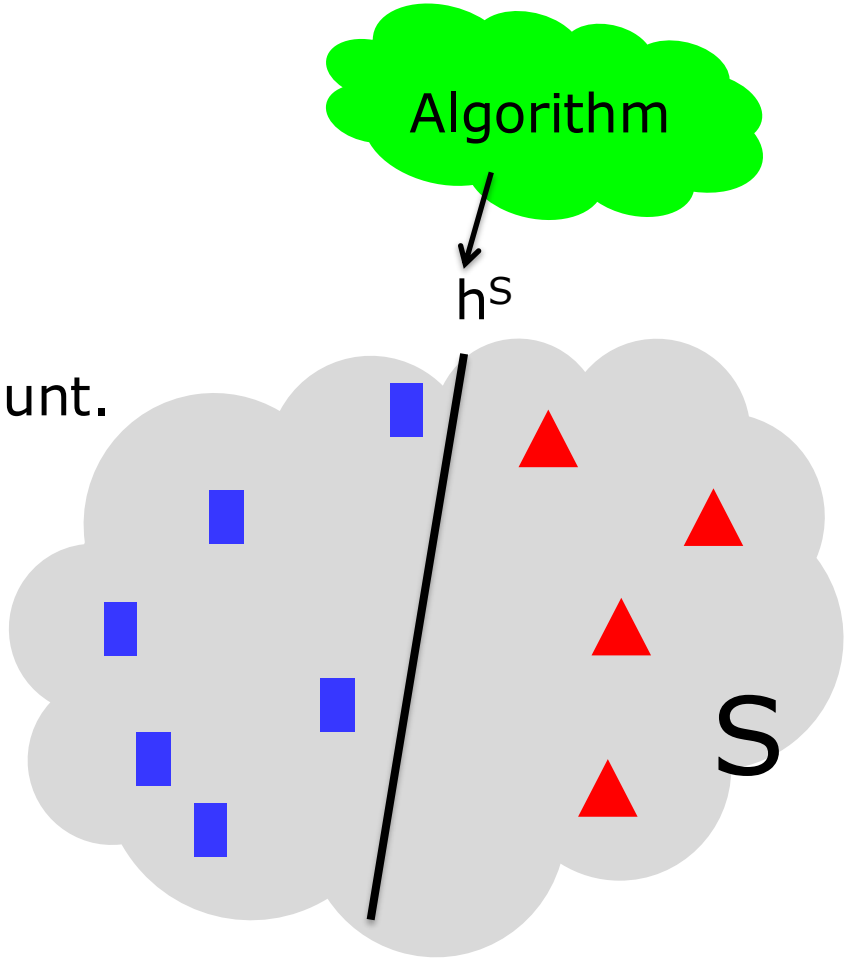


- Here h^S is the hypothesis produced by learning algorithm on data set S.

- Bounding expected out-of-sample error of hypothesis produced by learning algorithm on the random data set S.

Bias Variance

- Bias-Variance:
  - Bound E_{out}.
  - Take learning algorithm into account.



- Formally we bound:
  - $\mathbb{E}_S[E_{out}(h^S)]$
  - Depends on algorithm.

- VC-bound comparison:
  - $Pr_S[\exists\, h : |E_{in}(h) - E_{out}(h)| > \varepsilon] \leq 8((2n)^d+1)e^{-\varepsilon^2 n/8}$
  - Independent of algorithm.

Bias Variance

- Bounding $\mathbb{E}_S[E_{out}(h^S)]$:
  - $\mathbb{E}_S[E_{out}(h^S)] =$
  - $\mathbb{E}_S[\mathbb{E}_x[(h^S(x)-f(x))^2]] =$
  - $\mathbb{E}_x[\mathbb{E}_S[(h^S(x)-f(x))^2]] =$
  - $\mathbb{E}_x[\mathbb{E}_S[h^S(x)^2] + f(x)^2 - 2f(x)\mathbb{E}_S[h^S(x)]]$.

f is independent of S.

- Start by examining $\mathbb{E}_S[h^S(x)]$.

Bias Variance

- Examining $\mathbb{E}_S[h^S(x)]$.

- Intuition:
  - Repeatedly sample data sets $S_1, S_2, \ldots, S_m$ and train hypothesis h^{Si} for each.
  - Then $\frac{1}{m}\sum_i h^{Si}(x)$ tends to $\mathbb{E}_S[h^S(x)]$.

- Intuively an "average" hypothesis $\bar{h}(x) = \mathbb{E}_S[h^S(x)]$.

- The average hypothesis $\bar{h}(x)$ is what you get if you sample infinitely many S_i's in experiment above.

Bias Variance

- The average hypothesis $\bar{h}(x) = \mathbb{E}_S[h^S(x)]$:

- Example:
  - Unknown target function $f(x)=x^2 - x + 3$.
  - Data set: 2 uniform random points in [-10,10], i.e.
  - $(x_1, x_1^2 - x_1+3)$ and $(x_2, x_2^2 - x_2+3)$.

- Learning algorithm:
  - Return line through points (linear regression).

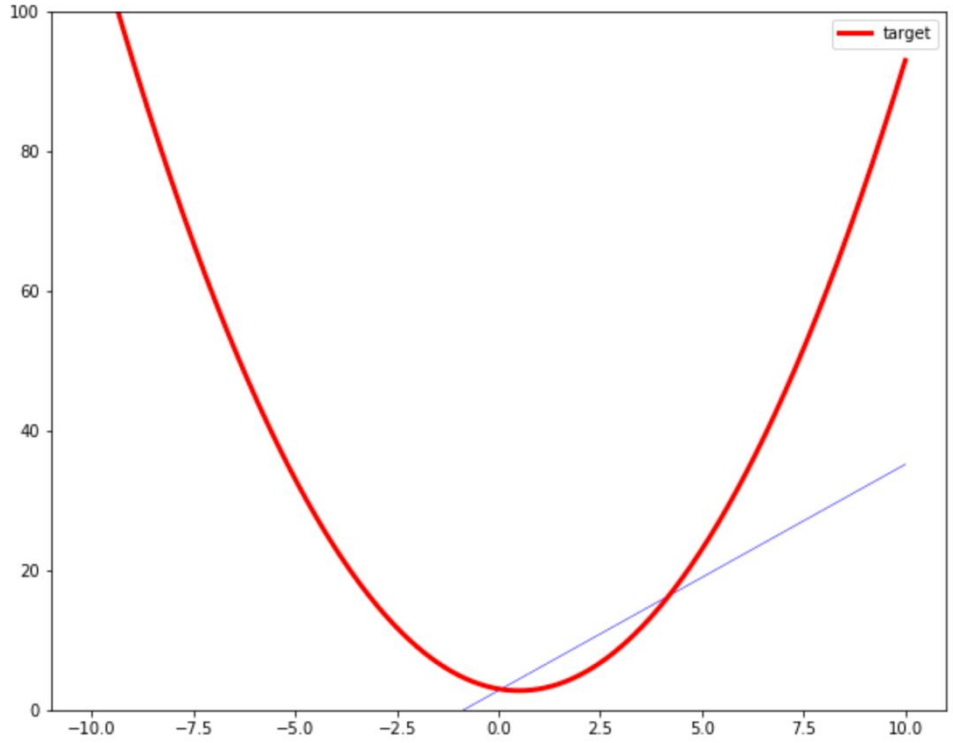
- What is the average hypothesis $\bar{h}(x)$?

Bias Variance

- The average hypothesis $\bar{h}(x) = \mathbb{E}_S[h^S(x)]$:

- Example:
  - Target $f(x)=x^2 - x + 3$.
  - Data set 2 points.
- Algorithm:
  - Fit line.

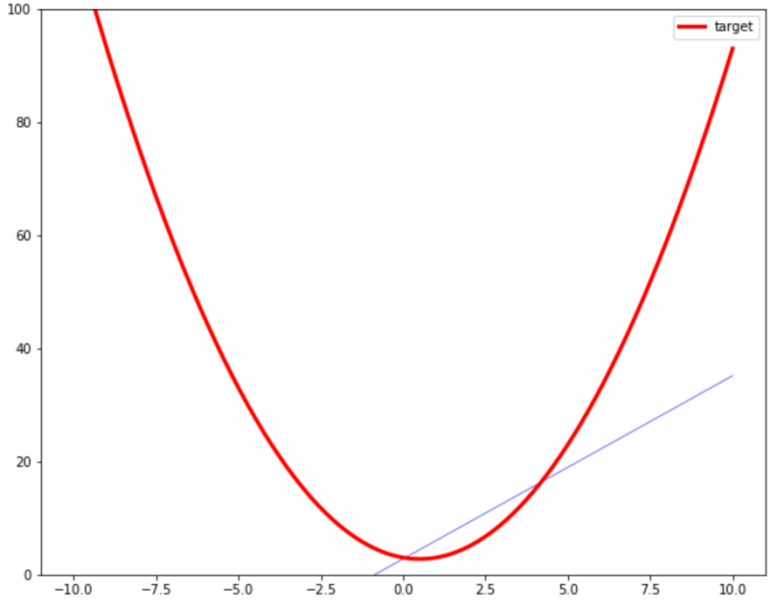
- What is the average hypothesis $\bar{h}(x)$?



Bias Variance

- $\bar{h}(x)$
- Line through (y=ax+b):
- $(x_1, x_1^2 - x_1 + 3)$ and $(x_2, x_2^2 - x_2 + 3)$.

- Slope:
- $a = (x_2^2 - x_2 + 3 - x_1^2 + x_1 - 3)/(x_2 - x_1)$
- $a = (x_2^2 - x_2 - x_1^2 + x_1)/(x_2 - x_1)$
- $a = (x_2 + x_1 - 1)(x_2 - x_1)/(x_2 - x_1)$
- $a = x_2 + x_1 - 1$

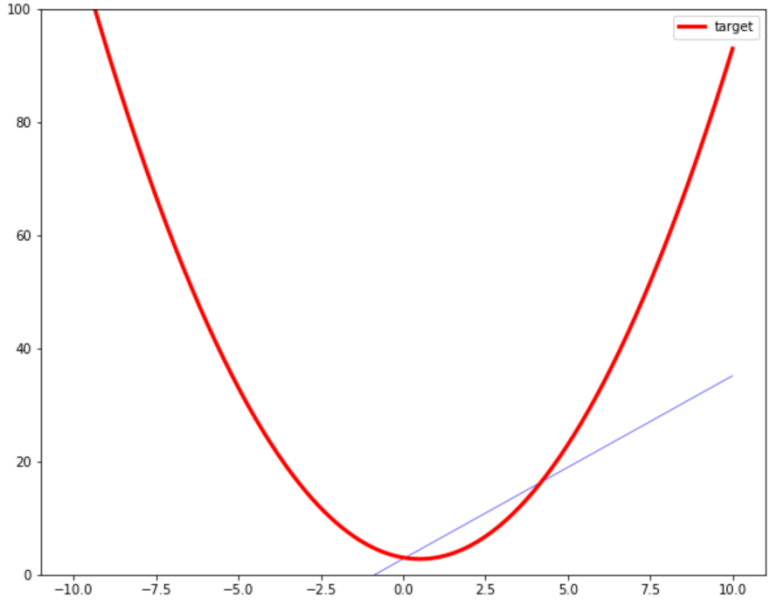


Bias Variance

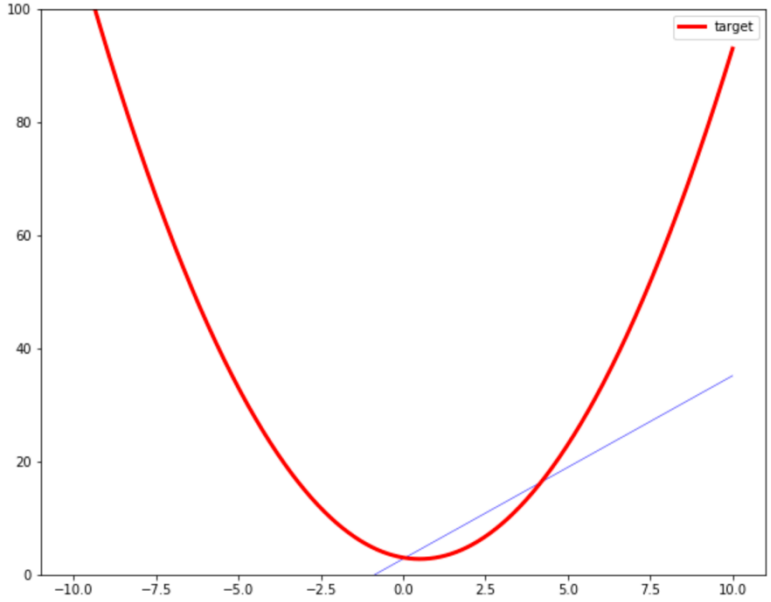
- $\bar{h}(x)$
- Line through (y=ax+b):
- $(x_1, x_1^2 - x_1 + 3)$ and $(x_2, x_2^2 - x_2 + 3)$.

- Slope:
- $a = x_2 + x_1 - 1$

- Offset:
- $b = x_1^2 - x_1 + 3 - x_1 a$
- $b = x_1^2 - x_1 + 3 - x_1 x_2 - x_1^2 + x_1$
- $b = 3 - x_1 x_2$



Bias Variance



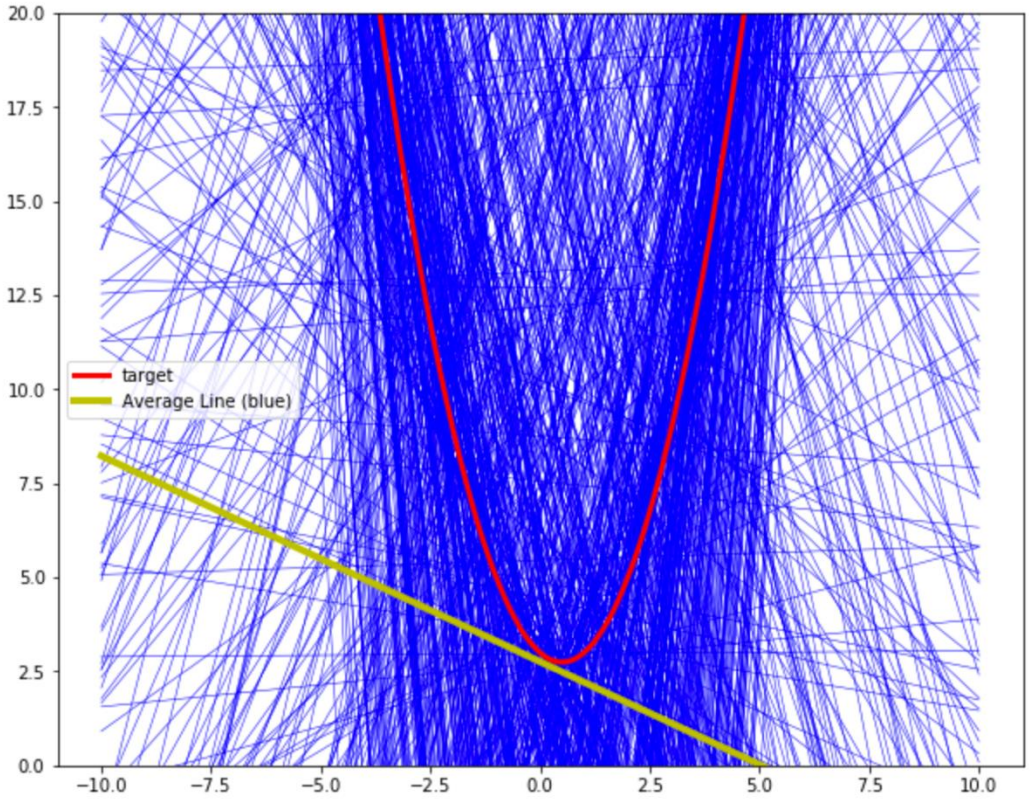
- $\bar{h}(x)$
- Line through (y=ax+b):
- $(x_1, x_1^2 - x_1 + 3)$ and $(x_2, x_2^2 - x_2 + 3)$.

- $a = x_2 + x_1 - 1$
- $b = 3 - x_1 x_2$

- Average hypothesis:
- $\bar{h}(x) = \mathbb{E}_S[h^S(x)] = \mathbb{E}_S[ax+b] = \mathbb{E}_S[a]x + \mathbb{E}_S[b]$
- $= \mathbb{E}_S[x_2 + x_1 - 1]x + \mathbb{E}_S[3 - x_1 x_2]$
- $= (\mathbb{E}_S[x_2] + \mathbb{E}_S[x_1] - 1)x + 3 - \mathbb{E}_S[x_1 x_2]$
- $= -x + 3 - \mathbb{E}_S[x_1]\mathbb{E}_S[x_2]$
- $= -x + 3.$

Bias Variance

- $\bar{h}(x) = -x + 3.$



notebook experiment

Bias Variance

- Bounding $\mathbb{E}_S[E_{out}(h^S)]$:
  - $\mathbb{E}_S[E_{out}(h^S)] =$
  - $\mathbb{E}_x[\mathbb{E}_S[h^S(x)^2] + f(x)^2 - 2f(x)\mathbb{E}_S[h^S(x)]]$.
- Start by examining $\mathbb{E}_S[h^S(x)]$.
- Average hypothesis $\bar{h}(x) = \mathbb{E}_S[h^S(x)]$.
- So:
  - $\mathbb{E}_S[E_{out}(h^S)] =$
  - $\mathbb{E}_x[\mathbb{E}_S[h^S(x)^2] + f(x)^2 - 2f(x)\bar{h}(x)]$.

Bias Variance

- Bounding $\mathbb{E}_S[E_{out}(h^S)]$:

Add 0.

  - $\mathbb{E}_S[E_{out}(h^S)] =$
  - $\mathbb{E}_x[\mathbb{E}_S[h^S(x)^2] + f(x)^2 - 2f(x)\bar{h}(x)] =$
  - $\mathbb{E}_x[\mathbb{E}_S[h^S(x)^2] - \bar{h}(x)^2 + \bar{h}(x)^2 + f(x)^2 - 2f(x)\bar{h}(x)] =$
  - $\mathbb{E}_x[\mathbb{E}_S[h^S(x)^2 - \bar{h}(x)^2]] + \mathbb{E}_x[(\bar{h}(x) - f(x))^2]$.

- Focus on $\mathbb{E}_S[h^S(x)^2 - \bar{h}(x)^2]$:
- Notice that:
  - $\mathbb{E}_S[(h^S(x) - \bar{h}(x))^2] =$
  - $\mathbb{E}_S[h^S(x)^2] + \bar{h}(x)^2 - 2\mathbb{E}_S[h^S(x)]\bar{h}(x) =$
  - $\mathbb{E}_S[h^S(x)^2] + \bar{h}(x)^2 - 2\bar{h}(x)^2 =$
  - $\mathbb{E}_S[h^S(x)^2 - \bar{h}(x)^2]$

Bias Variance

- Bounding $\mathbb{E}_S[E_{out}(h^S)]$:
  - $\mathbb{E}_S[E_{out}(h^S)] =$
  - $\mathbb{E}_x[\mathbb{E}_S[(h^S(x) - \bar{h}(x))^2]] + \mathbb{E}_x[(\bar{h}(x) - f(x))^2]$.
- So what does this mean?
- Observation:
  - $\mathbb{E}_x[(\bar{h}(x) - f(x))^2]$ is exactly $E_{out}(\bar{h})$.
  - The performance of the average hypothesis.
  - Called the **bias**.
    - **bias** = $\mathbb{E}_x[(\bar{h}(x) - f(x))^2]$

Bias Variance

- Bounding $\mathbb{E}_S[E_{out}(h^S)]$:
  - $\mathbb{E}_S[E_{out}(h^S)] =$
  - $\mathbb{E}_x[\mathbb{E}_S[(h^S(x) - \bar{h}(x))^2]] + \mathbb{E}_x[(\bar{h}(x) - f(x))^2]$.
- So what does this mean?
- Observation:
  - $\mathbb{E}_x[\mathbb{E}_S[(h^S(x) - \bar{h}(x))^2]]$ measures how far off the hypothesis h^S is from the average model in expectation.
  - Called the **variance**.
    - **variance** = $\mathbb{E}_x[\mathbb{E}_S[(h^S(x) - \bar{h}(x))^2]]$

Bias Variance

- The bias-variance decomposition:
  - $\mathbb{E}_S[E_{out}(h^S)]$ = **bias** + **variance**.

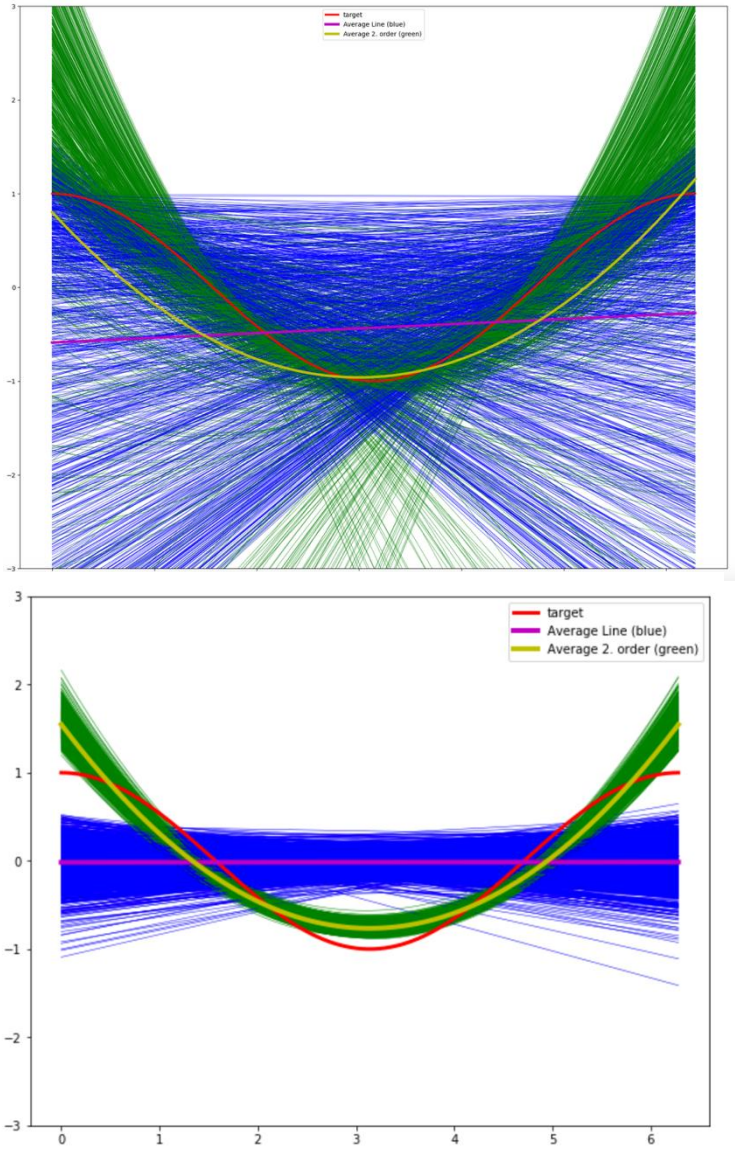
- Where:
  - **bias** = $\mathbb{E}_x[(\bar{h}(x) - f(x))^2]$
  - **variance** = $\mathbb{E}_x[\mathbb{E}_S[(h^S(x) - \bar{h}(x))^2]]$

- Variance intuitively measures how ”wildly” the learned hypothesis varies on changes in training data.
- High variance means prone to overfitting noise.
- Intuition: Complex models have low bias, but high variance.
- Simple models have high bias, but low variance.
- More data reduces variance.

Bias Variance

- The bias-variance decomposition:
  - $\mathbb{E}_S[E_{out}(h^S)]$ = **bias** + **variance**.

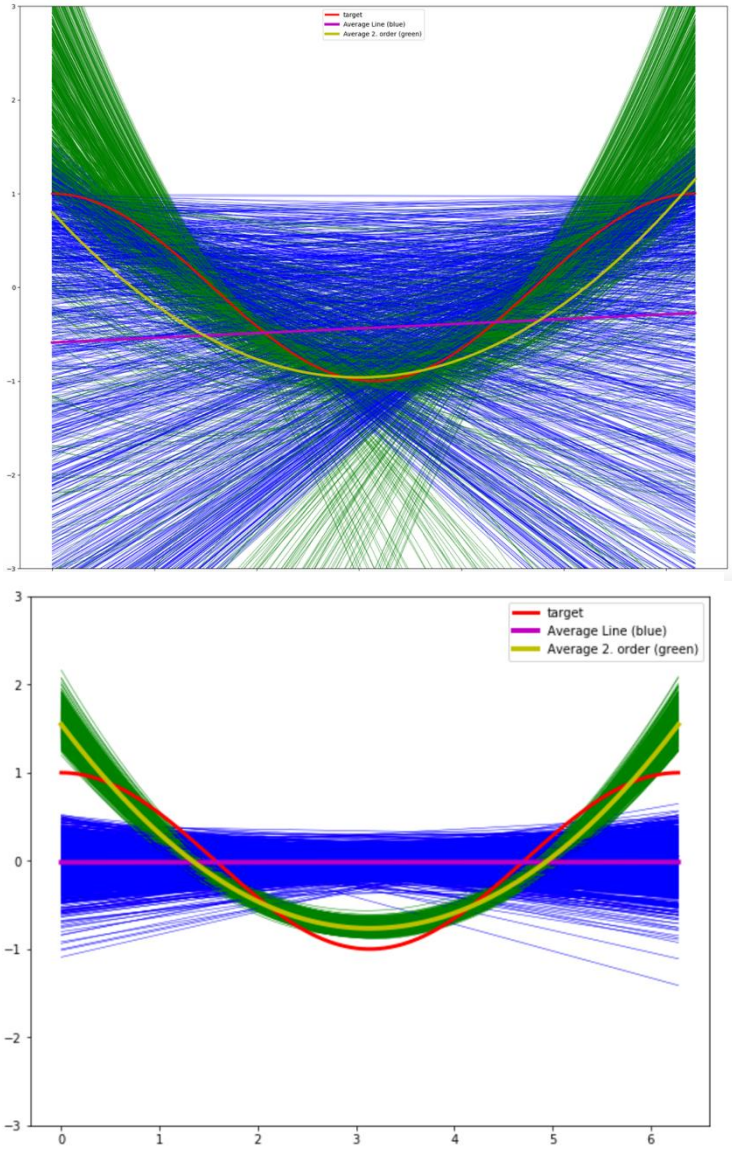
- Example:
  - Fitting cos(x) from samples.
  - Linear model, vs. 2nd deg. poly.
  - 3 data points, vs. 40 data points.



notebook experiment

Bias Variance

- The bias-variance decomposition:
  - $\mathbb{E}_S[E_{out}(h^S)]$ = **bias** + **variance**.
- To improve learning:
  - Reduce either bias or variance.
  - Bias hard to do much more about.
  - Several ideas for reducing variance.
    variance = $\mathbb{E}_x[\mathbb{E}_S[(h^S(x) - \bar{h}(x))^2]]$
- Intuitively:
  - More stability in the returned hypothesis.



Bagging and Random Forest

Bagging

Bagging

- Bootstrap Aggregating
  - Technique for reducing variance in returned hypothesis.
  - Applies to any learning algorithm.
  - Gives most improvements in ”unstable” algorithms (high variance).
  - Especially useful for decision/regression trees.

Bagging

- Bootstrap Aggregating

- Some intuition:
  - Want to reduce variance.
  - Let X be random variable with variance σ^2.
  - Let $X_1,\ldots,X_m$ be i.i.d. and have same distribution as X.
  - Let $Y = (1/m)(X_1+X_2+\ldots+X_m)$ be average.
  - Then $Var(Y) = \sigma^2/m$.

- Averaging i.i.d. random variables reduces variance.

Bagging

- Bootstrap Aggregating

- Averaging i.i.d. random variables reduces variance.
- Bias-Variance decomposition:
  - $\mathbb{E}_S[E_{out}(h^S)]$ = **bias** + **variance**.
- with
  - **variance** = $\mathbb{E}_x[\mathbb{E}_S[(h^S(x) - \bar{h}(x))^2]]$.

- Idea:
  - If we could somehow train hypotheses $h_1,\ldots,h_m$ on m i.i.d. data sets $S_1,\ldots,S_m$ then we can use average hypothesis:
  - $h^*(x) = (1/m)(h_1(x)+\ldots+h_m(x))$.

Bagging

- Bootstrap Aggregating

- Idea:
  - If we could somehow train hypotheses $h_1,\ldots,h_m$ on m i.i.d. data sets $S_1,\ldots,S_m$ then we can use average hypothesis:
  - $h^*(x) = (1/m)(h_1(x)+\ldots+h_m(x))$.
  - Would reduce variance.

- Problem:
  - Do not have m data sets.
  - Partitioning training set into m chunks reduces amount of data for training. So worse bias.

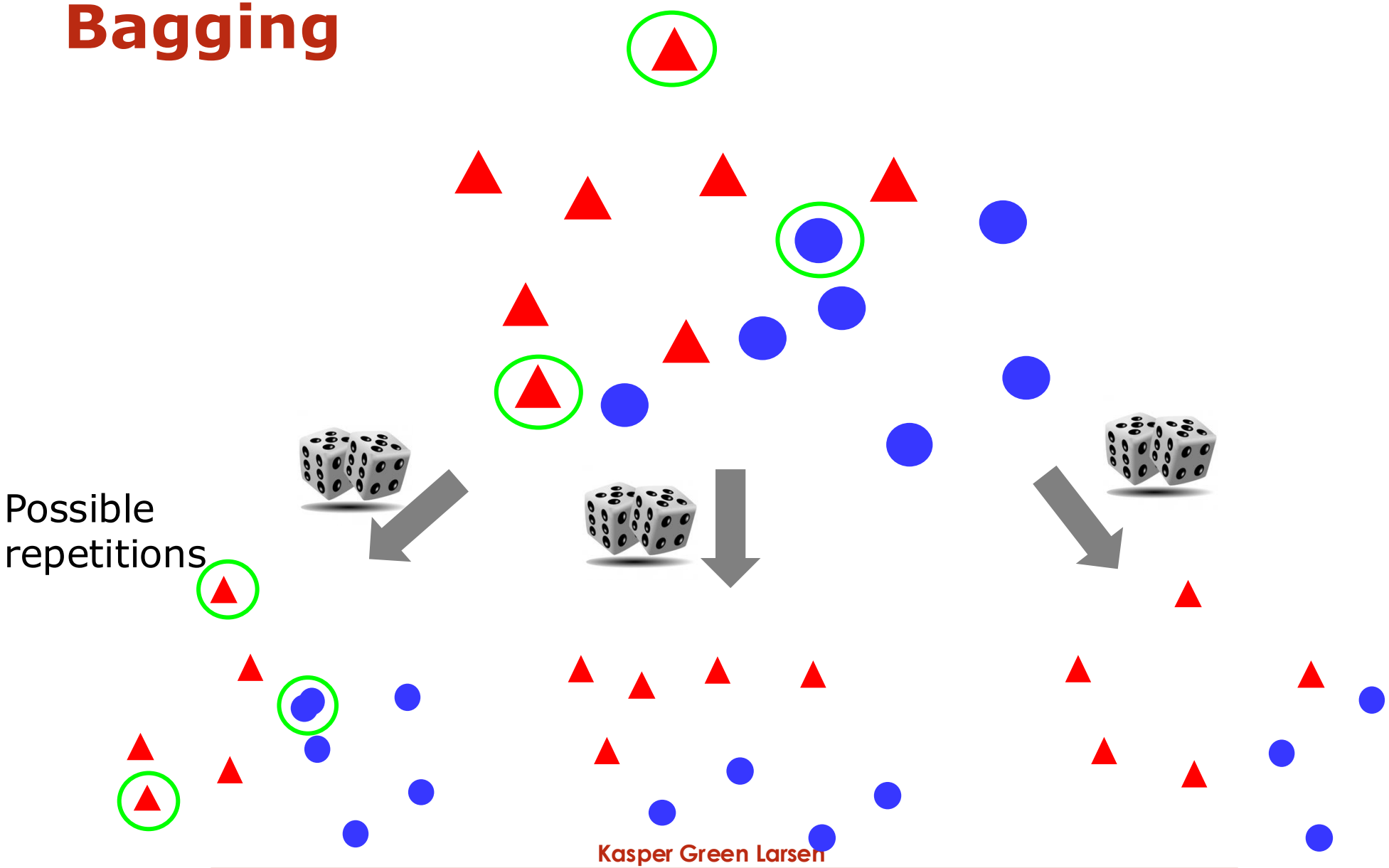
Bagging

- Bootstrap Aggregating

- Bagging:
  - Do m times:
    - Sample n' < n data points from S (with replacement). Call set S_i.
    - Train hypothesis h_i on S_i.
  - Output $h = (1/m)(h_1+\ldots+h_m)$ for regression.
  - Output $h = \text{sign}(h_1+\ldots+h_m)$ for classification.

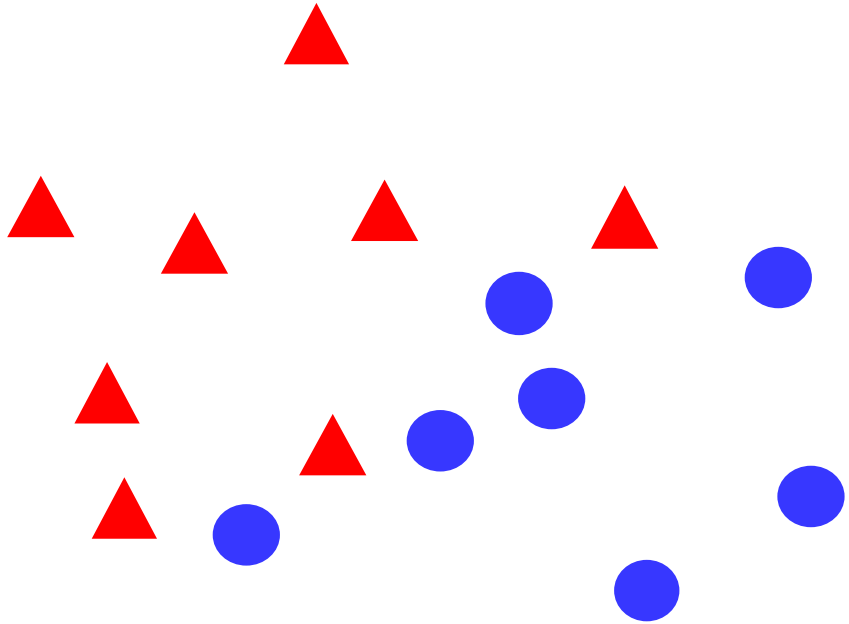
- Observations:
  - Reduces risk of overfitting noise on a point.
  - Should reduce variance.

Bagging

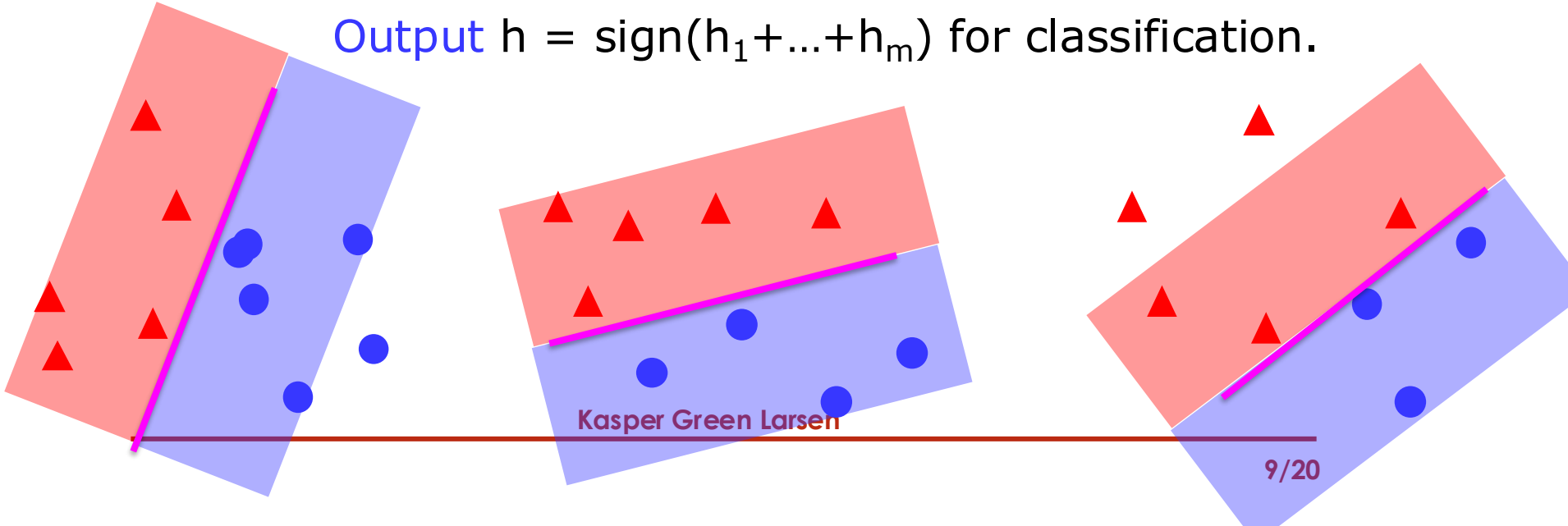


Bagging

Train hypothesis h_i on S_i.

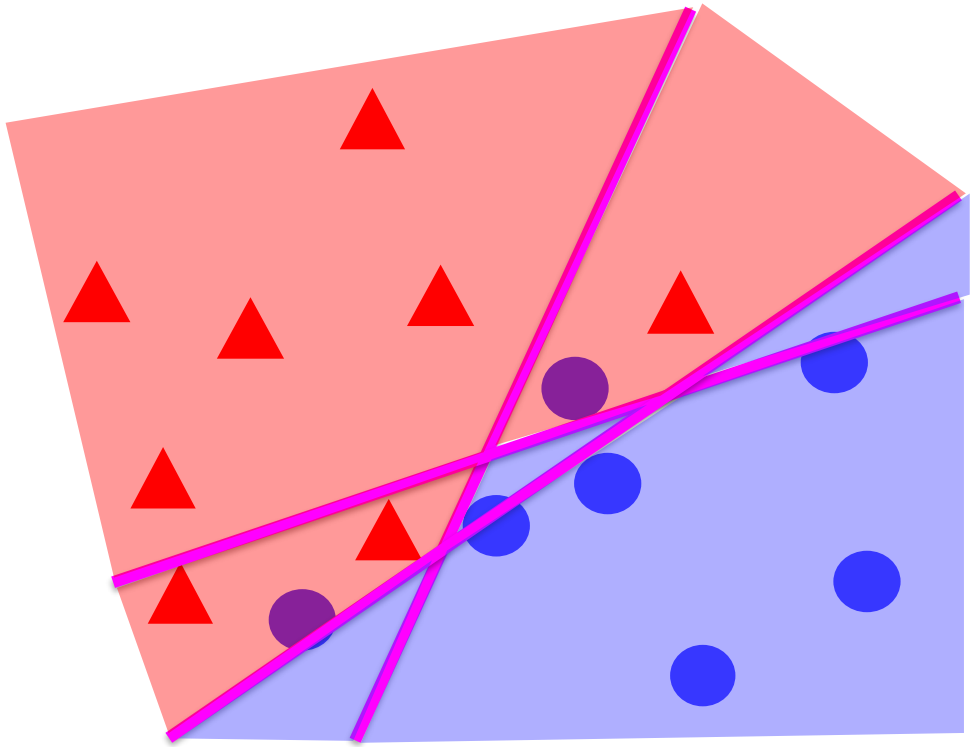


Output $h = \text{sign}(h_1+\ldots+h_m)$ for classification.

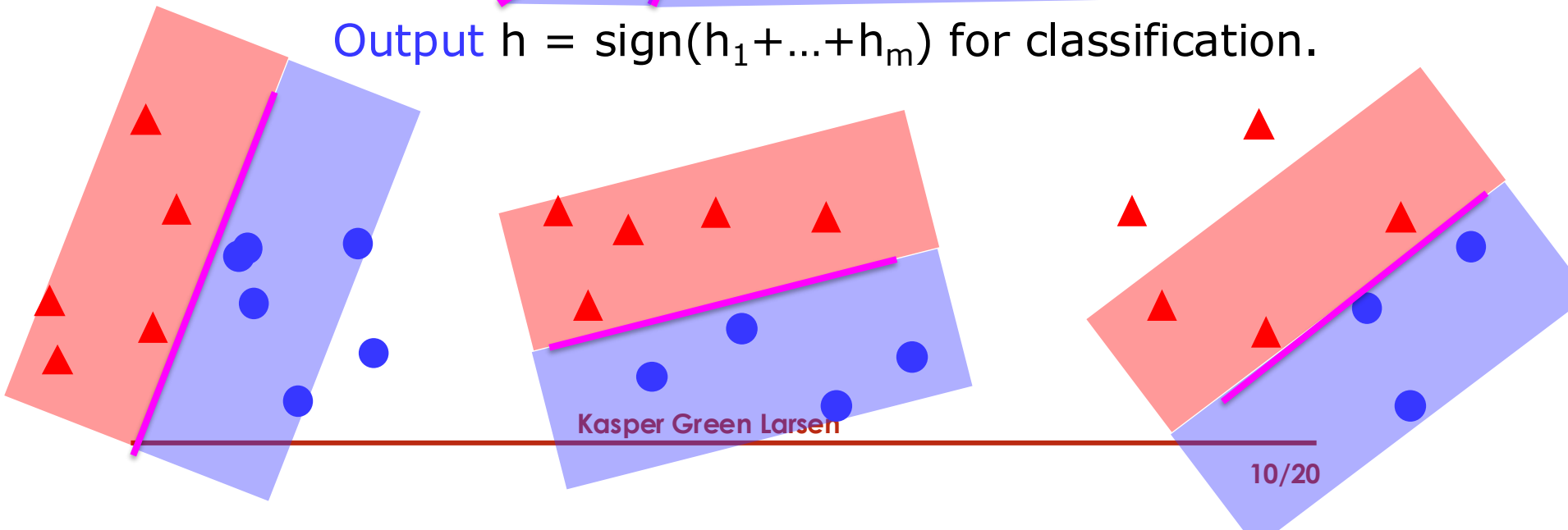


Bagging

Train hypothesis h_i on S_i.



Output $h = \text{sign}(h_1+\ldots+h_m)$ for classification.



Random Forest

Random Forest

- Recall Bagging Intuition:

- Some intuition:
  - Want to reduce variance.
  - Let X be random variable with variance σ^2.
  - Let $X_1,\ldots,X_m$ be i.i.d. and have same distribution as X.
  - Let $Y = (1/m)(X_1+X_2+\ldots+X_m)$ be average.
  - Then $Var(Y) = \sigma^2/m$.

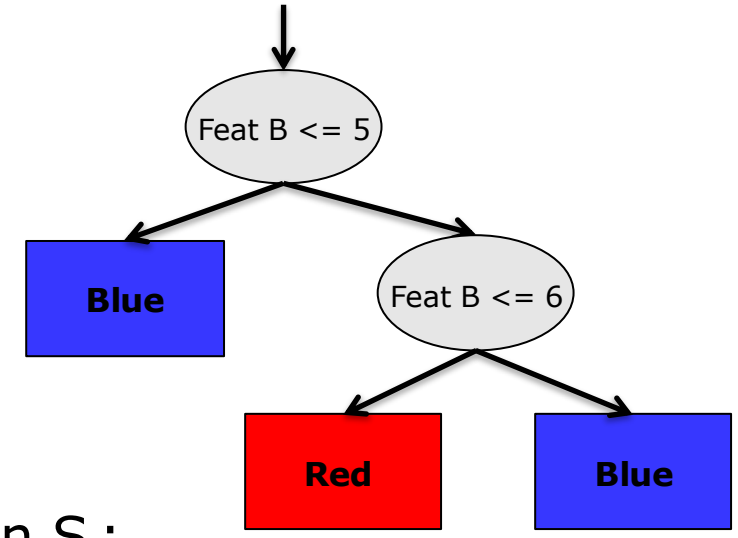
- Averaging i.i.d. random variables reduces variance.
- Bagging: Train on many subsamples of same data set and avrg.
- But the hypotheses trained on subsamples of same data set are not independent (they are i.d. though).

Random Forest

- Random Forest Intuition:
  - Let $X_1,\ldots,X_m$ be i.d. with variance σ^2.
  - Assume pairwise correlation $\rho>0$.
  - Let $Y = (1/m)(X_1+X_2+\ldots+X_m)$ be average.
  - Then $\mathrm{Var}(Y) = \rho\sigma^2 +(1- \rho)\sigma^2/m$.
- We will not prove this.

- Idea:
  - Try to reduce the correlation between the hypotheses trained on subsampled data sets.
  - Applied to decision and regression trees (hence the name).

Random Forest



- Random Forest:
  - Do m times:
    - Sample n' < n data points from S.
    - Call set S_i.
    - Train decision/regression tree T_i on S_i:
      - Each time computing a split:
        - Randomly choose p of the d data features.
        - Only allow splitting based on these p features.
  - Output $h = (1/m)(T_1+\ldots+T_m)$ for regression.
  - Output $h = \text{sign}(T_1+\ldots+T_m)$ for classification.

- Suggestions:
  - Classification: $p=\sqrt{d}$, Regression: $p=d/3$.
  - Or: Grid Search!

Out-of-Bag Samples

- Out-of-Bag Samples:
  - For both Random Forest and Bagging, we have data not used for training in each iteration.
  - Perform validation using these data points!
  - Saves a validation set / automatic cross validation.

Experiment

- Decision Tree vs. Bagging vs. Random Forest

- Data uniform in $[0, 2]^{15}$
- Target function: $f((x_1,\ldots,x_{15})) = (x_1-1)^2+\ldots+(x_{15}-1)^2$
- Want to compare bias and variance.

- How can we compute bias and variance?
  - Cheat and approximate them.

Experiment

- Decision Tree vs. Bagging vs. Random Forest

- Data uniform in $[0, 2]^{15}$
- Target function: $f((x_1,\ldots,x_{15})) = (x_1-1)^2+\ldots+(x_{15}-1)^2$
- Want to compare bias and variance.

- How can we compute bias and variance?
  - **bias** = $\mathbb{E}_x[(\bar{h}(x) - f(x))^2]$
  - **variance** = $\mathbb{E}_x[\mathbb{E}_S[(h^S(x) - \bar{h}(x))^2]]$

Experiment

- Estimating **bias**:
  - $\mathbb{E}_x[(\bar{h}(x) - f(x))^2] =$
  - $\mathbb{E}_x[(\mathbb{E}_S[h^S(x)]-f(x))^2]$.

- Sample a large test set (X,y).
- Sample many data sets S_i.
- For each S_i, train model h^{Si}.
- Evaluate h^{Si} on X to obtain predictions y^{Si}
- Average all predictions y^{Si} to obtain $y' \approx \bar{h}(X)$
- Bias is estimated by $(1/|X|)(\|y-y'\|^2)$.

Experiment

- Estimating **variance**:
  - $\mathbb{E}_x[\mathbb{E}_S[(h^S(x) - \bar{h}(x))^2]]$
- A Hack:
  - $\mathbb{E}_S[E_{out}(h^S)]$ = **bias** + **variance**.
- Need only estimate $\mathbb{E}_S[E_{out}(h^S)]$.
- How:
  - Sample many data sets S.
  - For each, train model h^S.
  - Evaluate h^S on large test set to approximate $E_{out}(h^S)$.
  - Average over all the sampled data sets S.

Experiment

- Decision Tree vs. Bagging vs. Random Forest

- Data uniform in $[0, 2]^{15}$
- Target function: $f((x_1,\ldots,x_{15})) = (x_1-1)^2+\ldots+(x_{15}-1)^2$

```
Decision Tree eout bias variance 2.50446987835079 - 1.1158177884332001 - 1.38865208991759
Bagging eout bias variance 1.4739702377407626 - 1.206091498788218 - 0.26787873895254455
Random Forest eout bias variance 1.4788389242387183 - 1.2080366159899174 - 0.2708023082488009
```